Distributed Systems

About Midterm

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Possible Topics for Midterm

- Distributed Systems
- Lamport's "Happened Before" Relation
- Logical Clocks vs Vector Clocks
- Rules to Implement Vector Clocks
- Properties of Vector Clocks
- Chandy's "Snapshot" Algorithm
- Ordering Events with Vector Clocks
- Balance and Tradeoff among Communication, Processing, and Storage
- Speedup

Possible Topics for Midterm

- Layered vs Middleware Communication Approaches
- Communication Types
- RMI Components
- RPC Steps
- Asynchronous RPC
- Passing Value Parameters in RPC
- Connection-Oriented Socket Communication
- Using a Buffer to Reduce Jitter
- Interleaved Communication

Possible Topics (Cont'd)

- Memory Consistency Models
- Memory Coherence
- DSM vs Message Passing
- False Sharing
- Choosing the DSM Page Size
- Page Synch Methods (Update Options)
- Copy Set (in Memory Coherence)
- Dynamic Page Ownership Approaches
- Lazy vs Eager Release Consistency
- Location Transparency vs Independence

Possible Topics (Cont'd)

- Sharing Semantics
- Choosing the Cache Unit Size
- Cached File Modification Policies
- Stateful vs Stateless Service
- DFS/AFS Scalability
- Large Cluster vs SM Machines
- Data-parallel vs Pipelined-parallel DNN (Deep Neural Network) Training
- RDMA (Remote Direct Memory Access)
- CXL (Compute eXpress Link)

Logical Clocks [Lamport78]

- Ways of Assigning Numbers to Events

$$LC_i(e), \text{where } e \text{ is an event in process } p_i$$

- Clock (Correctness) Condition

$$e \rightarrow e^* \quad \Rightarrow \quad LC(e) < LC(e^*)$$

  - Satisfied if the following two conditions hold:

$$(1)\ k < l \quad \Rightarrow \quad LC(e_i^k) < LC(e_i^l) \quad (i = 1, \ldots, N)$$

$$(2)\ e_i = send(m), e_j = receive(m) \ \ for\ Message\ m$$

$$\Rightarrow \quad LC(e_i) < LC(e_j)$$

Vector Clocks (Cont' d)

- Vector-Timestamp Comparisons

$$VC = VC^* \; iff \; VC[j] = VC^*[j] \; (j = 1, ..., N)$$
$$VC \leq VC^* \; iff \; VC[j] \leq VC^*[j] \; (j = 1, ..., N)$$
$$VC < VC^* \; iff \; VC \leq VC^* \wedge VC \neq VC^*$$

- Properties

$$e \rightarrow e^* \; iff \; VC(e) < VC(e^*)$$

Strong Clock Condition

  - Hints for the Proof

$$e \| e^* \Rightarrow \neg\{VC(e) \leq VC(e^*) \vee VC(e) \geq VC(e^*)\}$$
$$(Note\ That\ VC_i[j] \leq VC_j[j])$$